

Project Documentation Phase

Functional Specification Document (FSD): Auto Ticket Classification

Attribute	Project Information
Project Title	Auto Ticket Classification using Flow Designer
Phase / Stage	6. Project Documentation
Document Title	Functional Specification Document (FSD)
Platform & Scope	ServiceNow Utah / Vancouver / Washington DC (Global Scope)
Team ID	SWTID-2026-4988

1. Introduction

- Project Title: Auto Ticket Classification using Flow Designer
- Project Author / Team: ServiceNow Automators Team (ID: AGY-SNOW-2025)
- Platform: ServiceNow Enterprise Cloud (Global Application Scope)
- Document Purpose: This Functional Specification Document (FSD) provides complete technical, operational, and architectural documentation for the automated incident classification solution.

2. Project Overview & Business Value

- Core Purpose: Eliminate manual IT incident triage for educational helpdesks by automatically classifying incoming support tickets and notifying callers instantly upon record creation.
- Key Capabilities:
 - Automated incident auto-numbering with prefix "INC" (INC00500+).
 - Relational choice dependency between Category and Subcategory.
 - Real-time Flow Designer event trigger on record creation.
 - Natural-language keyword condition rules for Network, Hardware, Access, and Performance issues.
 - Dynamic email notification dispatched to the caller email address.
 - Portable deployment packaging via ServiceNow Local Update Sets exported as XML.

3. System Architecture & Component Design

The solution is partitioned into four distinct architectural layers:

1. UI Presentation Layer: Incident WorkFlow form configured via Form Designer to capture Caller and Short Description.
2. Data Layer: Custom table `u_incident_workflow` with relational references to `sys_user` and `sys_user_group`.
3. Business Logic Layer: Flow Designer flow "Auto Classify School IT Tickets" evaluating conditional branches.
4. Notification & Output Layer: System Logs > Emails engine dispatching outbound SMTP acknowledgments.

4. Setup & Configuration Instructions (Step-by-Step)

Step 1: Create New Update Set

- Navigate to All > System Update Sets > Local Update Sets.
- Click New button. Set Name = "Project Update Set", Application = Global, State = In progress.
- Click Related Links: "Make This My Current Set" so all subsequent changes are tracked.

Step 2: Create Custom Table

- Navigate to All > System Definition > Tables. Click New.
- Set Label = "Incident WorkFlow", Name auto-populates as `"u_incident_workflow"`. Uncheck "Create module".
- Open Controls section. Check "Auto Number", set Prefix = "INC", Starting Number = 500, Number of digits = 5. Save.

Step 3: Field Creation & Form Design

- Click Related Links: "Show Form". Open context menu > Configure > Form Design.
- Drag and drop required fields: Caller (Reference to `sys_user`), Category (Choice), Subcategory (Choice), Short Description (String), Description (String), State (Choice), Assigned Group (Reference to `sys_user_group`), Assigned To (Reference to `sys_user`).
- Configure Choice lists for Category (Network, Hardware, Access, Performance) and State (New, In progress, On hold, Resolved, Closed).

Step 4: Configure Dependent Choices

- Right-click Subcategory field > Configure Dictionary.
- Click Related Links: "Advanced view". Open Dependent Field section.
- Check "Use dependent field" = true. Set "Dependent on field" = Category. Save.
- Under Choices section, assign dependent values: Wi-Fi -> network, Projector -> hardware, Forgot password -> access, Slow computer -> performance. Save.

Step 5: Build Flow Designer Automation

- Navigate to All > Process Automation > Flow Designer. Click New > Flow.

- Name: "Auto Classify School IT Tickets", Application: Global. Click Build Flow.
- Add Trigger: Record Created on Table [u_incident_workflow] where [Category is empty].
- Add Condition 1: If Short description contains "Wi-Fi" or "Network" -> Action: Update Record (Category = Network, Subcategory = Wi-Fi).
- Add Condition 2: Else If Short description contains "Projector" -> Action: Update Record (Category = Hardware, Subcategory = Projector).
- Add Condition 3: Else If Short description contains "Forgot password" -> Action: Update Record (Category = Access, Subcategory = Forgot Password).
- Add Condition 4: Else If Short description contains "Slow computer" -> Action: Update Record (Category = Performance, Subcategory = Slow Compter).
- Add Action: Send Email. Target Record = Trigger Record, To = Trigger > Caller > Email, Subject = "Your Request for the issue has been submitted.", Body = Ticket confirmation greeting. Click Done.
- Click "Activate" in top right corner.

Step 6: Export Update Set to XML

- Navigate to System Update Sets > Local Update Sets > Open "Project Update Set".
- Change State from "In progress" to "Complete". Click Save.
- Click Related Links: "Export to XML". The file "sys_remote_update_set_...xml" downloads to local disk.

5. Data Dictionary Reference

Field Label	Field Name	Type	Reference / Choices	Mandatory
Number	u_number	Auto Number	Prefix: INC (5 digits)	Read-Only (Auto)
Caller	u_caller	Reference	sys_user	Yes
Category	u_category	Choice	Network, Hardware, Access, Performance	No (Auto-classified)
Subcategory	u_subcategory	Choice	Wi-Fi, Projector, Forgot Password, Slow Compter	No (Auto-classified)
Short Description	u_short_description	String	Free text string (255)	Yes
Description	u_description	String	Multiline string	No
State	u_state	Choice	New, In progress, On hold, Resolved, Closed	No
Assigned Group	u_assigned_group	Reference	sys_user_group	No
Assigned To	u_assigned_to	Reference	sys_user	No

6. Known Edge Cases & Future Enhancements

- Edge Case Handling: When an incoming ticket does not contain any of the four designated keywords, the flow finishes cleanly without populating Category, routing the ticket to the human triage queue.
- Future Enhancement 1: Natural Language Processing / Predictive Intelligence machine learning classification for complex multi-sentence requests.
- Future Enhancement 2: Automatic technician assignment based on assignment group round-robin rules.
- Future Enhancement 3: Service Level Agreement (SLA) workflows measuring MTTA and MTTR with automated escalation triggers.